

# EE5507

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- CMOS process
  - Single transistor amplifier
  - CMOS reference
  - OTA design – dc analysis
  - OTA design – ac analysis
  - OTA design – high performance opamp
- OTA is fundamental building block
    - op amp in ADC and DAC also
    - OTA and op amp are different
      - OTA is operational transconductance amplifier
        - op amp is small output resistance (from output buffer stage, 10s  $\Omega$ ), to drive low resistance load
        - transconductance means the output stage is 10k-100k $\Omega$  output
          - so will need high resistor
            - 1M and 100k for inverting with gain -10
          - it is good for driving capacitive load
            - so can use 10 pF and 1 pF and get gain -10

|            |      |
|------------|------|
| Assignment | 20%  |
| Project    | 40%  |
| Final Exam | 40%  |
| Total      | 100% |

- Part 1: 3 assignment 1 project
- Part 2: 3-4 assignment 1 project
- Final Exam
  - 4 Q, 2 part 1, 2 part 2
  - open book, open PC just no wifi
    - can bring journal !!!

- Construct an opamp according to spec
- Schematic, design calculation, simulation, reasoning
- Cadence Virtuoso with gpdk045nm

- PDK has model, DRC rule, LVS rule
- Generic PDK is from cadence for learning
- report always show
  - vds and vdssat so can check for saturation
  - label must be big and easy to see

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### Cadence workshop

- stability

- Monte carlo for offset
- switch cap (sample and hold)
- in the weekends

<https://pollev.com/ay2122s2hch>

## **PART 2**

# **Recap/Review/Overview**

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## **1. Part 1 lecture material**

- Basic circuits, building blocks
- Single transistor amplifiers, OPA, OTA, Bandgap Reference

## **2. Part 2 lecture material**

- Higher level building blocks
- Abstraction concept
  - Eg. Active filters, data converters use OPA
- Practical circuits for useful functions in an IC
  - ADC, DAC, Comparator, Oscillators (Clock), S/H (Signal processing)
- Understanding of Noise

- Week 6 (17 Feb) Data Converters
  - Tutorial/HW4 assign 18 Feb due 14 March
- \* **Recess Week**
- Week 7 (3 March) Analog Filters (1) \* 2<sup>nd</sup> half of class
  - Project assign 4 March due 14 April
- Week 8 (10 March) Analog Filters (2)
- Week 9 (17 March) Noise
  - Tutorial/HW5 assign 18 March due 24 March
- Week 10 (24 March) Comparators, Sample & Hold
  - Tutorial/HW5 assign 25 March due 3 April
- Week 11 (31 March) HOLIDAY
- Week 12 (7 April) Oscillators
  - Tutorial/HW6 assign 10 April due 16 April
- Week 13 (14 April) Industry talk \* Review/Overflow
  - Project Presentations

EE5507 – Analog Integrated Circuit  
Design (Part 2)

- Tutorial/HW
  - 3 accounting for 10% of final grade
  - For your learning as you work through problems
  - **No late submissions**
  - Solutions provided
- Project
  - 1 accounting for 20% of final grade
  - 15 minute (max) physical presentation + Q & A (week 13). Presentation slots can be booked but first come first serve
  - Student groups of 2 (**no passengers or free riders!**)
  - Since it clashes with semester end, strongly suggest you schedule time (**time management is important life skill**)
  - Follow marking rubric